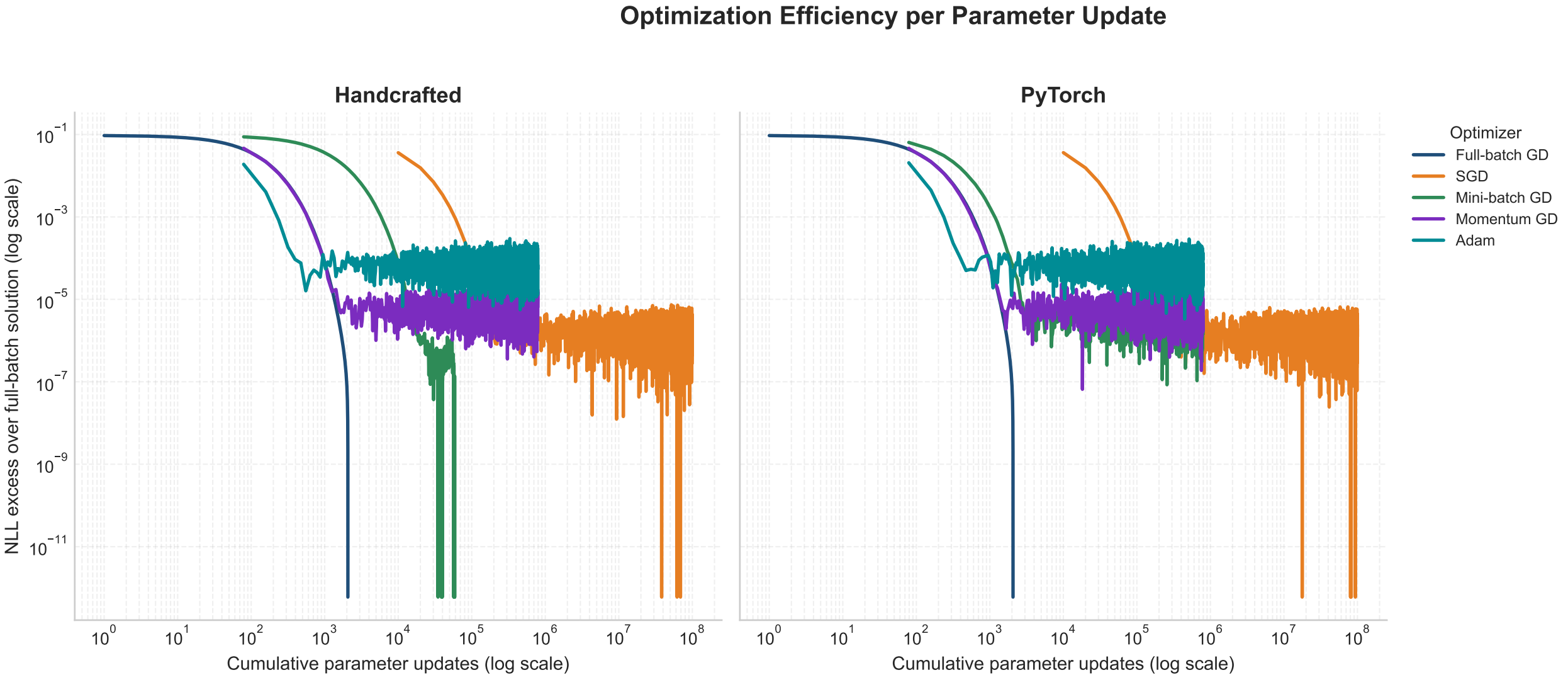

Optimization Efficiency per Parameter Update
Handcrafted
PyTorch
NLL excess over full-batch solution (log scale)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{0} 10^{1} 10^{2} 10^{3} 10^{4} 10^{5} 10^{6} 10^{7} 10^{8}
Cumulative parameter updates (log scale)
Optimizer
Full-batch GD
SGD
Mini-batch GD
Momentum GD
Adam